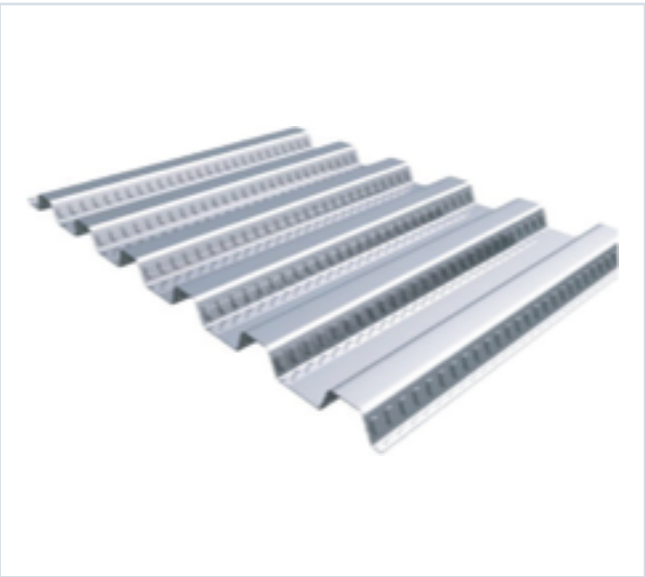


DECKING PRODUCT SUBMITTAL

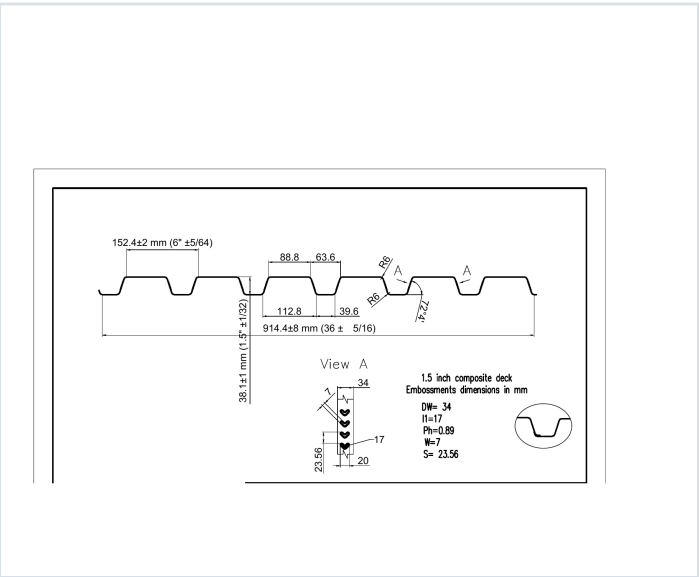
15CD-FY40-22-G40

PRODUCT	GRADE	GAUGE	COATING
1.5" Composite Deck	Grade 40	22 ga	G40

PRODUCT VIEW



PROFILE DRAWING



SECTION PROPERTIES · 22 GAUGE

Thickness (in.)	Weight (psf)	Fy (ksi)	S+ (in ³ /ft)	S- (in ³ /ft)	M+ (in-lb/ft)	M- (in-lb/ft)	I+ (in ⁴ /ft)	I- (in ⁴ /ft)
0.0295	1.6	40	0.173	0.184	4135	4415	0.147	0.171

SOURCE AND USE

1. Source: Refined Metal Steel Catalog 2025, published pages 41, 42, 43, 44
2. ASD section values above are the exact published row for this grade and gauge.
3. Coating selection does not change the published structural performance values.
4. Verify project-specific loading, fastening, span, concrete, and code requirements with the design professional.

PUBLISHED PERFORMANCE DATA

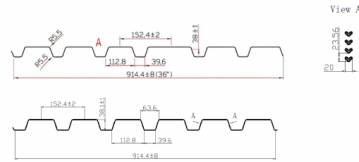
15CD-FY40-22-G40 · 22 GAUGE

OFFICIAL STEEL CATALOG EXCERPTS

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STEEL CATALOG PAGE 41

GRADE 40 STEEL



Section Properties

Gage	Design Thickness (inches)	Weight (pcf)	F, ksi	S _x • (in ³) per foot	S _y • (in ³) per foot	ASD (φ = 1.67)		I _x • (in ⁴) per ft.	I _y • (in ⁴) per ft.
						M / d in.-lbs. per foot	M / d in.-lbs. per foot		
22	0.0096	16	40	0.173	0.094	4335	4405	0.147	0.071
20	0.0098	20	40	0.219	0.231	5246	5533	0.187	0.216
18	0.0074	26	40	0.299	0.332	7164	7473	0.263	0.290
16	0.0098	30	40	0.383	0.390	9166	9333	0.350	0.363

Note: All section properties and ASD flexural strengths are calculated in accordance with ANSI/S3C RD-2017, ANSI S308-2017 and ANSI S308-2016.

Shear and Web Crippling

Gage	V _u (lb/ft)	Web Crippling (R _w), lb/ft One Flange Loading End Bearing			Web Crippling (R _w), lb/ft One Flange Loading Interior Bearing		
		1-1/2"	2"	3"	1-1/2"	2"	3"
22	1939	640	704	850	877	951	1076
20	3042	96	1002	1149	1284	1388	1561
18	4005	153	1670	1902	2218	2386	2667
16	4975	2345	2547	2895	3476	3723	4139

Note: All section properties and ASD flexural strengths are calculated in accordance with ANSI/S3C RD-2017, ANSI S308-2017 and ANSI S308-2016.

Allowable Uniform Downward Loads, ASD (PSF)

Span	Gage	5'-0"	5'-6"	6'-0"	6'-6"	7'-0"	7'-6"	8'-0"	8'-6"	9'-0"	9'-6"	10'-0"
Single	22	110	98	77	65	56	49	43	38	34	31	28
	20	140	116	97	83	71	62	55	48	43	39	35
	18	158	128	102	85	73	64	56	49	44	40	36
	16	244	202	170	145	125	109	95	85	75	68	63
Double	22	168	147	122	102	87	75	66	58	51	46	41
	20	348	282	232	192	162	142	125	109	95	85	75
	18	399	323	262	212	182	162	142	125	109	95	85
	16	599	499	402	332	282	242	212	182	162	142	125
Triple	22	147	122	102	87	75	66	57	51	45	41	37
	20	384	312	258	218	184	162	142	125	109	95	85
	18	449	368	302	252	212	182	162	142	125	109	95
	16	661	542	442	362	302	252	212	182	162	142	125

Note: 1. All section properties and ASD (d = 1/8") uniform loads are calculated in accordance with ANSI/S3C RD-2017, ANSI S308-2017 and ANSI S308-2016.
2. Loads shown in table are uniformly distributed superimposed loads in psf. Span length assumes center-to-center spacing of supports. Tabulated loads shall not be increased by assuming clear span conditions.
3. Bending Moment formula used for fixed-end stress limitations are: Single and Two Span: $M = \frac{wL^2}{8}$; Three Span or More: $M = \frac{wL^2}{16}$
4. Web crippling and clear have not been accounted for in these tables. Required bearing should be determined based on specific span conditions.

STEEL CATALOG PAGE 42

Span	Gage	5'-0"	5'-6"	6'-0"	6'-6"	7'-0"	7'-6"	8'-0"	8'-6"	9'-0"	9'-6"	10'-0"
Single	22	77	58	45	35	28	23	19	16	13	11	10
	20	98	74	57	45	36	29	24	20	17	14	12
	18	109	80	60	45	36	30	24	20	17	14	12
	16	138	102	76	57	45	36	30	24	20	17	14
Double	22	168	147	122	102	87	75	66	58	51	46	41
	20	348	282	232	192	162	142	125	109	95	85	75
	18	399	323	262	212	182	162	142	125	109	95	85
	16	599	499	402	332	282	242	212	182	162	142	125
Triple	22	147	122	102	87	75	66	57	51	45	41	37
	20	384	312	258	218	184	162	142	125	109	95	85
	18	449	368	302	252	212	182	162	142	125	109	95
	16	661	542	442	362	302	252	212	182	162	142	125

Note: For loads that cause L/120 Deflection, multiply by 2.0. For loads that cause L/180 Deflection, multiply by 1.5. For loads that cause L/360 Deflection, multiply by 0.67.

Construction Span Table – 20 psf Construction Load

Total Slab Depth	Deck Type	Maximum Unshored Clear Span
350 (14-200) 31 PSF	15x10x22 ga	5'-4" 6'-0" 6'-6"
	15x10x18 ga	6'-0" 7'-0" 7'-6"
	15x10x16 ga	6'-6" 7'-6" 8'-0"
	15x10x14 ga	7'-0" 8'-0" 8'-6"
400 (14-250) 37 PSF	15x10x22 ga	5'-4" 6'-0" 6'-6"
	15x10x20 ga	6'-0" 7'-0" 7'-6"
	15x10x18 ga	7'-0" 8'-0" 8'-6"
	15x10x16 ga	7'-6" 8'-6" 9'-0"
450 (14-300) 43 PSF	15x10x22 ga	5'-4" 6'-0" 6'-6"
	15x10x20 ga	6'-0" 7'-0" 7'-6"
	15x10x18 ga	7'-0" 8'-0" 8'-6"
	15x10x16 ga	7'-6" 8'-6" 9'-0"
500 (14-400) 49 PSF	15x10x22 ga	5'-4" 6'-0" 6'-6"
	15x10x20 ga	6'-0" 7'-0" 7'-6"
	15x10x18 ga	7'-0" 8'-0" 8'-6"
	15x10x16 ga	7'-6" 8'-6" 9'-0"
550 (14-500) 55 PSF	15x10x22 ga	5'-4" 6'-0" 6'-6"
	15x10x20 ga	6'-0" 7'-0" 7'-6"
	15x10x18 ga	7'-0" 8'-0" 8'-6"
	15x10x16 ga	7'-6" 8'-6" 9'-0"
600 (14-600) 61 PSF	15x10x22 ga	5'-4" 6'-0" 6'-6"
	15x10x20 ga	6'-0" 7'-0" 7'-6"
	15x10x18 ga	7'-0" 8'-0" 8'-6"
	15x10x16 ga	7'-6" 8'-6" 9'-0"

Note: Web crippling and clear have not been accounted for in these tables. Required bearing should be determined based on specific span conditions.

PUBLISHED PERFORMANCE DATA

15CD-FY40-22-G40 · 22 GAUGE

OFFICIAL STEEL CATALOG EXCERPTS

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STEEL CATALOG PAGE 43

22 ga Normalweight Concrete (145 pcf, f'c = 3,000 psi)									
Slab Thickness (inches)	Weight (psf)	5'-0"	5'-6"	6'-0"	6'-6"	7'-0"	7'-6"	8'-0"	
3.5	31	400	354	295	249	212	182	158	
4	37	400	400	372	334	298	231	200	
4.5	43	400	400	400	383	326	289	244	
5	49	400	400	400	400	387	331	290	
5.5	55	400	400	400	400	400	387	336	
6	61	400	400	400	400	400	400	383	
Slab Thickness (inches)		8'-6"	9'-0"	9'-6"	10'-0"	10'-6"	11'-0"	11'-6"	12'-0"
3.5	138	121	107	95	84	75	67	60	
4	175	154	135	120	107	96	86	77	
4.5	215	188	166	147	131	118	106	95	
5	254	225	197	175	156	142	128	115	
5.5	294	259	229	204	180	163	148	132	
6	335	296	262	233	208	186	168	151	

20/18/16 ga Normalweight Concrete (145 pcf, f'c = 3,000 psi)									
Slab Thickness (inches)	Weight (psf)	5'-0"	5'-6"	6'-0"	6'-6"	7'-0"	7'-6"	8'-0"	
3.5	31	400	400	359	313	259	223	194	
4	37	400	400	400	383	328	283	248	
4.5	43	400	400	400	400	400	345	310	
5	49	400	400	400	400	400	400	357	
5.5	55	400	400	400	400	400	400	400	
6	61	400	400	400	400	400	400	400	
Slab Thickness (inches)		8'-6"	9'-0"	9'-6"	10'-0"	10'-6"	11'-0"	11'-6"	12'-0"
3.5	170	150	132	118	105	94	85	76	
4	215	190	168	150	134	120	108	98	
4.5	263	232	206	183	164	147	133	120	
5	313	276	245	218	195	176	158	143	
5.5	364	321	285	254	227	204	184	167	
6	400	387	346	290	260	234	211	191	

Note
In case of the profile of the embossments, there is no gain in strength for the composite deck-slab when the deck gets thicker than 20 gauge. However, the construction spans do get longer for 18 and 16 gauge deck.

STEEL CATALOG PAGE 44

20/18/16 ga Normalweight Concrete (145 pcf, f'c = 3,000 psi)									
Slab Thickness (inches)	Weight (psf)	5'-0"	5'-6"	6'-0"	6'-6"	7'-0"	7'-6"	8'-0"	
3.5	23	400	341	285	241	206	178	154	
4	28	400	400	361	308	261	225	196	
4.5	33	400	400	400	372	319	275	239	
5	37	400	400	400	400	379	327	285	
5.5	42	400	400	400	400	400	380	332	
6	46	400	400	400	400	400	400	379	
Slab Thickness (inches)		8'-6"	9'-0"	9'-6"	10'-0"	10'-6"	11'-0"	11'-6"	12'-0"
3.5	135	119	106	94	84	75	68	61	
4	172	150	136	120	107	96	87	78	
4.5	210	185	164	146	131	118	106	96	
5	250	221	196	175	157	141	127	115	
5.5	289	257	228	204	182	164	148	134	
6	333	294	261	233	209	188	170	154	

20/18/16 ga Lightweight Concrete (115 pcf, f'c = 3,000 psi)									
Slab Thickness (inches)	Weight (psf)	5'-0"	5'-6"	6'-0"	6'-6"	7'-0"	7'-6"	8'-0"	
3.5	23	400	400	344	291	249	216	188	
4	28	400	400	400	378	321	274	238	
4.5	33	400	400	400	400	388	335	292	
5	37	400	400	400	400	400	400	349	
5.5	42	400	400	400	400	400	400	400	
6	46	400	400	400	400	400	400	400	
Slab Thickness (inches)		8'-6"	9'-0"	9'-6"	10'-0"	10'-6"	11'-0"	11'-6"	12'-0"
3.5	90	142	129	115	104	93	84	76	
4	201	185	165	147	132	118	107	97	
4.5	257	227	202	180	162	146	132	119	
5	306	271	241	216	194	174	158	145	
5.5	357	316	281	251	226	203	184	167	
6	400	362	322	288	259	235	211	192	

Note
In case of the profile of the embossments, there is no gain in strength for the composite deck-slab when the deck gets thicker than 20 gauge. However, the construction spans do get longer for 18 and 16 gauge deck.